SPECIFICATION DOCUMENT · V1.0.0

Web Application Design System & Interface Guidelines

Core Foundations & Philosophy

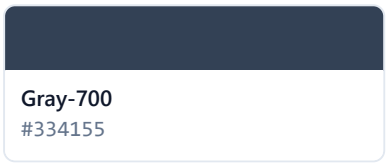
- **Clarity over cleverness:** Content takes precedence over decoration. Visual flair serves navigation, information scent, and hierarchy.
- **Density with purpose:** Compact tables and high-density layouts for data dashboards; breathable, open layouts for onboarding and consumer-facing flows.
- **Predictable affordance:** Standardized hover, active, and focus states across all interactive surfaces.

1. Color Palette & Token Architecture

Primitive neutrals establish background elevation and readable hierarchy, while semantic tokens designate actionable and stateful workflows.



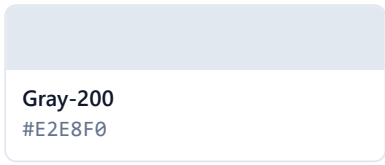
Gray-900
#0F172A



Gray-700
#334155



Gray-400
#94A3B8



Gray-200
#E2E8F0

Role	Light Hex	Dark Hex	Typical Usage
Primary (Brand)	`#2563EB`	`#3B82F6`	Primary action buttons, active tabs, focus rings
Primary Hover	`#1D4ED8`	`#60A5FA`	Hover state for brand actions
Success	`#16A34A`	`#22C55E`	Badges, confirmations, positive financial trends
Warning	`#D97706`	`#F59E0B`	Session timeouts, non-destructive alerts, caveats
Destructive / Error	`#DC2626`	`#EF4444`	Form errors, delete confirmations, failed syncs
Info	`#0284C7`	`#38BDF8`	Inline hints, feature announcements, tooltips

2. Typography & Scale

UI Stack: `Inter, -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, sans-serif`
Monospace Stack: `"JetBrains Mono", "SF Mono", Consolas, monospace`

Level	Size / Line Height	Weight	Usage
Display	32px / 40px	700 (Bold)	Marketing heroes, initial welcome screens
Heading 1 (H1)	24px / 32px	600 (Semi-Bold)	Main view and dashboard screen titles
Heading 2 (H2)	20px / 28px	600 (Semi-Bold)	Section divides, modal dialog headers
Heading 3 (H3)	16px / 24px	500 (Medium)	Card titles, grouped fieldset legends
Body Regular	14px / 20px	400 (Regular)	Default copy, inputs, table data cells
Body Small	12px / 16px	400 / 500	Form helper text, badges, secondary timestamps

3. Spacing, Grid & Depth

- `4px` (0.25rem): Icon-to-text micro gaps, compact tags.
- `8px` (0.50rem): Standard component gap, button and input vertical padding.
- `16px` (1.00rem): Card internal padding, standard form field margins.
- `24px` (1.50rem): Structural card margins, multi-column gutters.
- `32px` (2.00rem): Macro layout section spacing.

Elevation & Box Shadows

- **Flat (0px):** Page background canvas and flush flat panels.
- **Low / Card:** `0 1px 3px rgba(0,0,0,0.06)` — Standard content containers.
- **Mid / Dropdown:** `0 4px 6px -1px rgba(0,0,0,0.08), 0 2px 4px -2px rgba(0,0,0,0.04)` — Popovers, flyouts, and menus.
- **High / Modal:** `0 20px 25px -5px rgba(0,0,0,0.1), 0 8px 10px -6px rgba(0,0,0,0.05)` — Floating dialogs and command bars.

4. Component Specifications

Buttons

- **Primary:** Solid `#2563EB` with white text. Limit to at most one primary action per view section.
- **Secondary:** 1px solid `#E2E8F0` border, white background, `#0F172A` text.
- **Destructive:** Solid `#DC2626` or outlined red. Always require secondary confirmation for destructive actions.
- **Sizing:** 32px height for compact tables/toolbars; 40px height for default forms and views. Border radius: 6px.

Form Controls

- **Inputs:** Fixed height 40px, 1px border (`#E2E8F0`), 6px border radius, 14px text.
- **Focus State:** Border transitions to `#2563EB` with an outer glow: `box-shadow: 0 0 0 3px rgba(37, 99, 235, 0.2)`. Never remove focus outlines without providing this ring.
- **Error State:** Border transitions to `#DC2626` accompanied by persistent helper text placed below the input.

Modals & Overlays

- **Backdrop:** `rgba(15, 23, 42, 0.45)` with `backdrop-filter: blur(4px)`.
- **Standard Widths:** Small: 440px (confirmations), Medium: 600px (standard forms), Large: 880px (data inspectors).
- **Interaction:** Must bind the `Esc` key to close unless unsaved modifications exist in the form.

5. Accessibility & Motion Rules

- **Contrast Compliance:** Maintain a minimum ratio of **4.5:1** for normal UI text and **3:1** for graphical controls and larger text (WCAG AA).
- **Keyboard Navigation:** Modals must trap focus while open; form inputs must follow logical document order (`tabindex`).
- **Transitions:**
  - Micro-interactions (buttons, hovers): `150ms ease-out`
  - Drawers and sliding panels: `250ms cubic-bezier(0.16, 1, 0.3, 1)`
  - Full modal appearance: `200ms ease-in-out`
- **Reduced Motion:** Honor the `@media (prefers-reduced-motion: reduce)` query by collapsing transition durations to zero.